MIL-F-14614D(ME)
18 July 1979
SUPERSEDING
MIL-F-14614C(ME)
6 December 1974

MILITARY SPECIFICATION

FRAME AND TABLE ASSEMBLY, STEREOPLOTTER, PROJECTION: MULTIPLEX, 32-INCH LENGTH, WITH CHEST

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 32-inch multiplex frame with table, blower, voltage controls, and chest.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-843	- Cushioning, Material, Cellulosic.
PPP-T-60	- Tape: Packaging, Waterproof.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-10562	- Projector, Stereoplotter, Wide-Angle, Vertical, Multiplex Mapping Equipment (Set of Three Projectors in a Case with Accessories).
MIL-P-15024/6	- Plate, Identification.
MIL-S-15769	- Straightedges, Steel (Machinists' and Toolmakers').

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13213E3100	- Frame and Table Assembly, Stereoplotter, Projection: Multiplex, 32-Inch Length, with Chest.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The frame and table assembly and chest shall be in accordance with Top Assembly TA13213E3100 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the frame and table assembly. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced frame and table assembly). The contractor shall furnish one or more frame and table assemblies as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce frame and table assemblies that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.4 Electrical system. The electrical system shall operate from 120-volt, 60 Hertz (Hz), single-phase current.

3.4.1 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.4.1.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus- and moisture-resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.4.1.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.5 Identification marking. The frame and table assembly shall be identified in accordance with MIL-STD-130. The marking shall be applied to the frame and table assembly on plates conforming to MIL-P-15024/6. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

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3.6 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.4):

Item No.	Description	Identification	Quantity
1	Projector, Stereoplotter, Wide-Angle, Vertical, Multiplex Mapping Equipment (Set of Three Projectors in a Case with Accessories).	MIL-P-10562	1 set

3.7 Workmanship. Aluminum alloy used in fabrication of the equipment shall be free from kinks and sharp bends. The straightening of the material shall be done by methods that will not cause damage to the metal. The surface and edges of all finished members shall be free from sandholes, blowholes, shrinks, cracks, and scratches. Alloys up to 0.5 inch (13 millimeters (mm)) in thickness may be cut by shearing. All rough edges or burrs on sheared or sawed components shall be removed. The aluminum shall not be flame cut.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced frame and table assembly shall be cause for disassembly, in the presence of a Government representative, of the first produced frame and table assembly to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in

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detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced frame and table assembly. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-loaned property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced frame and table assembly inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced frame and table assembly inspection.

4.3.1 Examination. The first produced frame and table assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first produced frame and table assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each frame and table assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each frame and table assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The frame and table assembly shall be examined as specified herein for the following defects:

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101. Nonconformance to drawings.
102. Materials not as specified.
103. Components missing or not as specified.
104. Identification marking missing, incorrect, incomplete, or illegible.
105. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Surface plate flatness. A straightedge, not less than 5 feet (1524 mm) in length and conforming to MIL-S-15769, shall be used to check the flatness of the surface plate. The straightedge shall be used to check the following positions: Along a line 1 inch (25.4 mm) in from each edge, across each center line, and across each diagonal. Contact or lack of contact of the straightedge with the surface plate shall be checked along the entire contiguous length of the surface plate with feeler gages. A deviation more than 0.002 inch (0.05 mm) between contiguous surfaces of the straightedge and the plate surface shall constitute failure of this test.

4.5.2.2 Base bar rack operation. A set of three multiplex projectors conforming to MIL-P-10562 will be loaned by the Government for this test (see 3.6). The projector shall be placed on the base bar of the frame. The projector shall then be moved along the length of the frame using the horizontal direction screw to determine that the gears on the projector properly mesh with the rack on the base bar. Any improper meshing of the teeth, irregular movement, or binding shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. AQL shall be 2.5 percent defective.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

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106. Materials, methods, or containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
107. Surfaces requiring a preservative, not coated with preservative, or not wrapped or covered as specified for Level A.
108. Finished surface of the surface plate not protected with cushioning as specified for Level A.
109. Contents not placed in the chest as specified for Level A.
110. Chests not packed in an upright position.
111. Boxes not modified with skids as specified for Level A or Level B.
112. Strapping not as specified.
113. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unpainted exterior metal surfaces of components such as bolts, nuts, screws, base bar, or any other unpainted metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-9 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Coated surfaces of such components shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 2. The barrier material shall be secured in place with tape conforming to PPP-T-60, Type IV.

5.1.1.2 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1.

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5.1.1.3 Consolidation. Each complete frame and table assembly, and technical publications shall be consolidated in the chest. The finished surface of the surface plate shall be additionally protected with cushioning material conforming to PPP-C-843, Type II, Class C, to prevent damage.

5.1.2 Commercial. Each complete frame and table assembly, and technical publications shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Each chest containing a complete unit shall be packed with the top of the chest placed in an upright position in a close-fitting box conforming to PPP-B-601, Overseas Type, Style A or B, or PPP-B-621, Class 2, Style 2. The box shall be modified by the addition of skids in accordance with the box specification. Strapping shall be Finish A and placed adjacent to the skids.

5.2.2 Level B. Each chest containing a complete unit shall be packed as specified in 5.2.1, except boxes shall be Domestic Type or class as applicable.

5.2.3 Commercial. Each chest containing a complete unit shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The frame and table assembly is intended for use in conjunction with standard vertical and oblique multiplex equipment for the plotting of individual stereoscopic models.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced frame and table assembly, and number of frame and table assemblies required (see 3.2).
(c) When Government-loaned property is not required (see 3.6).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced frame and table assembly. Any changes or deviations of production frame and table assemblies from the approved first produced frame and table assembly during production will be subject to the approval of the contracting officer. Approval of the first produced frame and table assemblies will not relieve the contractor of his obligation to furnish frame and table assemblies conforming to this specification.

6.4 Government-loaned property. The contracting officer should arrange to lend the property specified in 3.6, when applicable (see 6.2).

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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Army - ME

Preparing activity:
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Review activities:
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DOCUMENT IDENTIFIER AND TITLE MIL-F-14614D(ME) Frame & Table Assy, Stereoplotter, Projection: Multiplex, 32-Inch Length, with Chest

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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